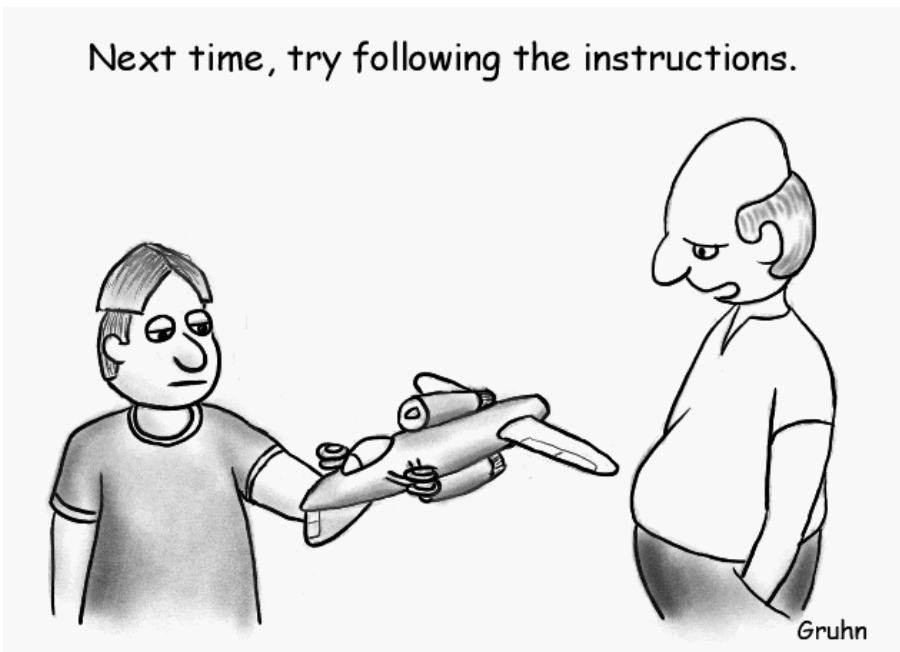
16

CASE STUDY

Chapter Highlights

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16.3 Case Description
16.4 Scope of Analysis
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16.1 Introduction

The case study presented in this chapter is intended to show how the material and techniques outlined in Chapters 1 to 15 can be used to specify, design, install, commission, and maintain a typical safety instrumented system (SIS). The intention is to cover the lifecycle steps presented in Chapter 2. By reviewing the solutions relating to specific issues in this case study, the reader will be able to further clarify the techniques, better understand and appreciate the need for good documentation, and resolve possible misunderstandings. This chapter should serve as an additional guide to those involved in specifying and designing safety systems.

The emphasis of this chapter is on the approach or method, *not* on the problem or the final solution. The case study and the documentation have been simplified so that the focus remains on "how to" rather than "how many."

A Word of Caution

The controls and protective instrumentation discussed in this chapter are oversimplifications offered for study purposes only. They do not in any way reflect what is or what may be required for actual installations. Also, the solutions and the system proposed do not necessarily comply with some internationally recognized standards (e.g., National Fire Protection Association, American Petroleum Institute).

16.2 The Safety Lifecycle and Its Importance

As highlighted in Chapter 2, the ANSI/ISA 84.00.01-2004, Parts 1-3 (IEC 61511 Mod) standard lists the various steps that need to be followed to ensure that safety instrumented systems are specified, designed, installed, and properly maintained. Figure 16.1 is a simplified version of the lifecycle. The various steps can be divided into three main phases:

1. risk analysis
2. design and implementation
3. operation and maintenance

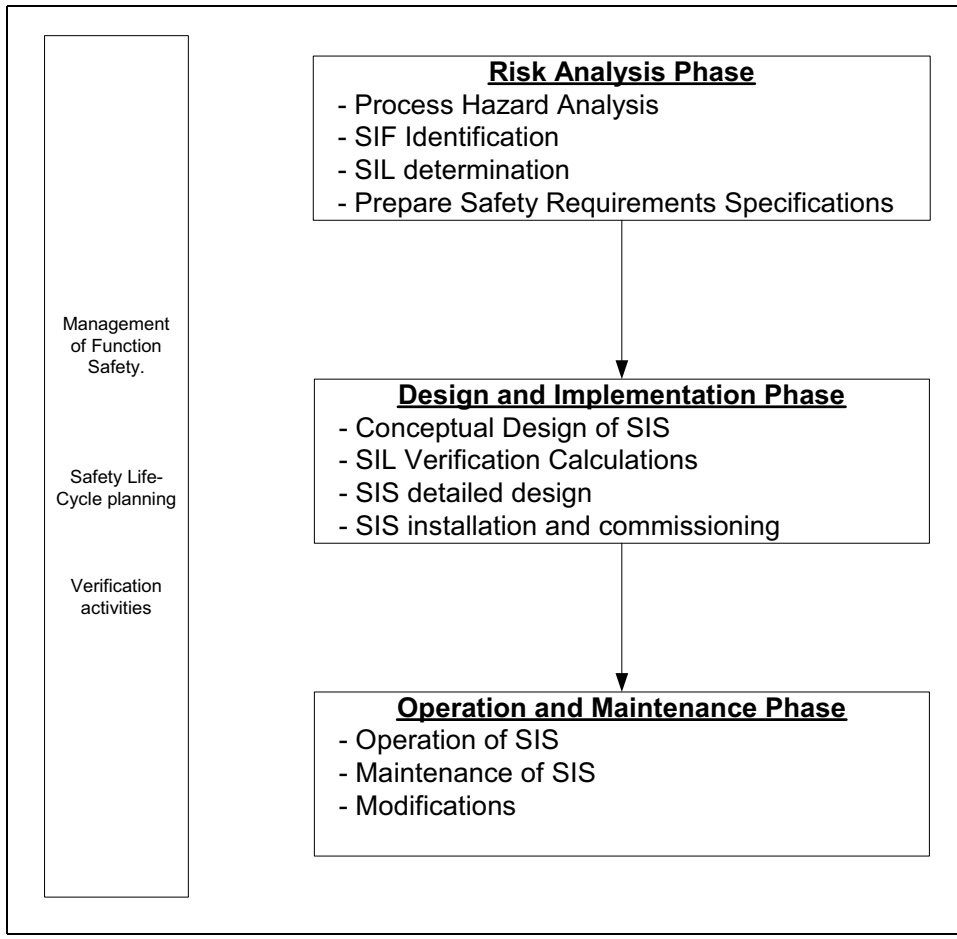


Figure 16-1: Safety Lifecycle Phases

The risk analysis phase covers the process hazards analysis, risk evaluation, identification of safety instrumented functions (SIFs), safety integrity level (SIL) determination, and the safety requirements specification (SRS).

The design and implementation phase covers the design and implementation of the SIS. This includes the SIL verification calculations to verify that the system being installed satisfies the SIL requirements, along with installation and commissioning.

The operation and maintenance phase deals with operating and maintaining the system throughout its life in order to ensure that the SIL for each function is sustained.

The standard also emphasizes that the management of functional safety, functional safety assessments, safety lifecycle structure and planning, and

verification activities should occur throughout the lifecycle. This is indicated by the vertical box on the left side of Figure 16-1.

All of the above activities need to be addressed for the following reasons:

- Chapter 2 reviewed the findings of the United Kingdom Health and Safety Executive (U.K. HSE) that showed that accidents due to control and safety system failures could be traced back to errors that could have been caught by adherence to such a lifecycle.
- Systematic failures can be reduced by following the lifecycle. Such failures are caused by inadequate specifications, design, software development, testing, training, and procedures.
- Safety integrity levels need to be recognized as a lifecycle issue. In order to sustain the SIL determined for the various safety functions, all of the lifecycle steps need to be completed and verified.

16.3 Case Description: Furnace/Fired Heater Safety Shutdown System

The safety instrumented system for the fuel controls of a furnace/fired heater, which is part of a crude unit in an oil refinery, needs to be upgraded. Plant management is concerned that the existing systems are inadequate, ineffective, and unreliable. The furnace is natural draft, natural gas fired, with continuous pilots. Figure 16-2 shows a diagram of the furnace and the associated basic process controls.

Description of Basic Process Controls

The feed flow to the furnace is controlled by regulatory control loops FC-9 and FC-10. A local self regulating pressure controller (PCV-6) controls the pilot gas pressure. The fuel gas flow and coil outlet temperature controls are part of a cascade control system. The gas flow controller (FC-3) is in cascade with temperature controller (TC-8). The flow controller manipulates valve FV-3 to control the gas flow at a set point manipulated by the temperature controller. All regulatory control loops are in the basic process control system (BPCS), which is located in a control room approximately 200 feet away from the furnace.

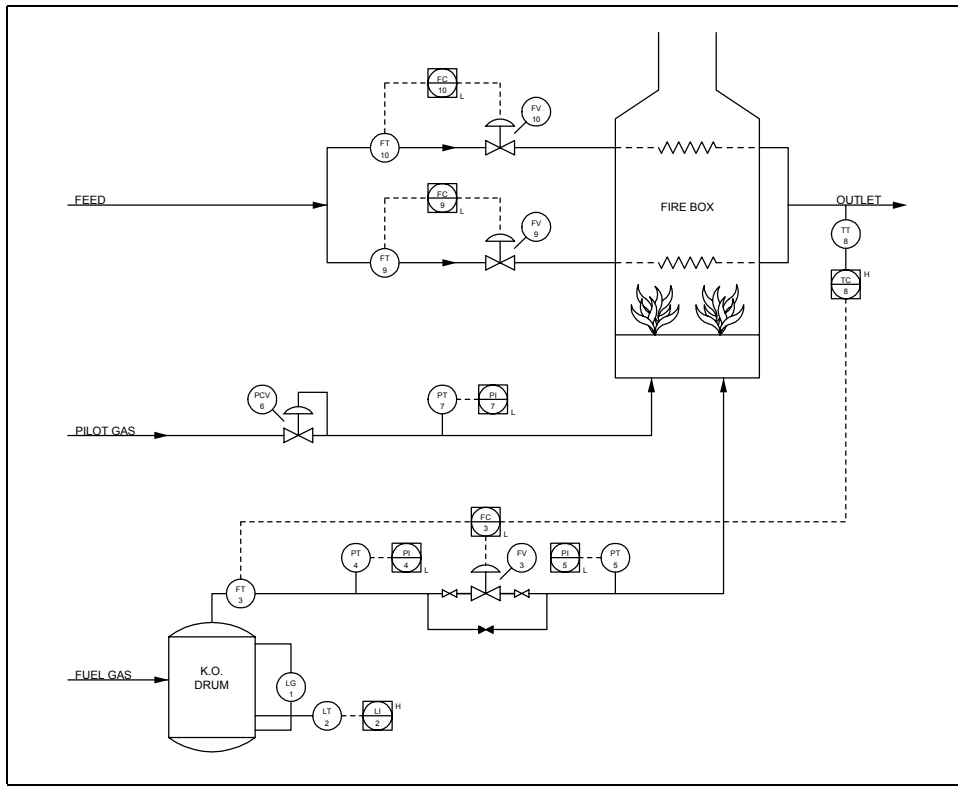


Figure 16-2: Fired Heater and Basic Process Controls

The following hazardous events were identified during the process hazard review:

Table 16-1: Furnace Hazards

Hazard	Possible Cause	Consequence	Likelihood
Furnace Explosion	Loss of pilots	Multiple Fatalities	Medium
Furnace fire	Loss of pilots, or loss of flow in furnace tube	$1,000,000 loss	Medium
Tube failure	Loss of flow in furnace tube	$1,000,000 loss	Medium

It was felt that no additional non-instrumented layers could be applied to prevent or mitigate the above events. The following safety and protective instrumentation was recommended.

1. Fuel gas to be shutoff on low flow on any pass (with 20-sec delay)
2. The fuel gas and pilots to be shutoff on low pilot gas pressure

3. A dedicated hardwired switch to be provided in the main control room for manual shutoff of fuel and pilots

A demand on each safety instrumented function is expected to occur at a frequency less than once per year. Also, one nuisance trip per year of the furnace due to a failure of the safety system can be tolerated. The cost impact of a nuisance trip is approximately $20,000.

16.4 Scope of Analysis

The scope of the analysis includes the following safety lifecycle steps:

1. define the target SIL for each SIF
2. develop the safety requirements specification (SRS)
3. SIS conceptual design
4. lifecycle cost analysis
5. verify that the conceptual design meets the SIL
6. detailed design
7. installation, commissioning, and pre-startup testing
8. establish operation and maintenance procedures

Not included in the above scope are other key lifecycle activities that also need to be completed prior to, or in conjunction with, the above activities. These are:

- Safety lifecycle planning. A safety lifecycle plan needs to be developed and established for the project. The standards do not require that operating companies follow the safety lifecycle as specifically outlined in the standards.

- Functional safety management activities. Some of these include:

    - creating a functional safety management plan
    - defining the roles and responsibilities of personnel
    - addressing personnel competency
    - documentation and documentation control
    - planning for functional safety verification
    - planning for functional safety assessment
    - safety audits

- Verification activities associated with each step of the lifecycle.

- Functional safety assessment. An assessment must at least be carried out prior to the startup of the plant.

16.5 Define Target SILs

A safety integrity level (SIL) must be determined for each safety instrumented function, several methods for SIL determination were outlined in Chapter 6, including:

Qualitative:Using the 3-D risk matrix

QualitativeUsing the risk graph

QuantitativeLayer of protection analysis (LOPA)

The 3-D risk matrix will be used for this case study. The consequence and likelihood of the hazardous events are documented in Table 16-1.

A Word of Caution

The consequence categories used in the SIL determination are the worst credible consequence assuming that no safeguards exist. The likelihood categories assume that all safeguards exist except the safety instrumented function (SIF) being reviewed. In many instances, the likelihood categories in the safety study and report *include* the SIF. If this is the case, the SIL team should redo the likelihood category assuming that the SIF does *not* exist.

The risk/SIL determination matrix shown below is a simplification of the matrix shown in Figure 6-2. This matrix shows the SIL required for various consequence and likelihood categories assuming no additional safety layers (as was determined during the safety study).

Consequence	**Hi**	2	3	3
	Med	2	2	3
	Low	1	2	2
		Low	**Med**	**Hi**
		Likelihood		

The SIL required for the two SIFs based on the above matrix is shown in Table 16-2.

Table 16-2: Safety Integrity Level Requirements for Fired Heater Example

Item	Safety Function	Consequence		Likelihood	SIL
1	Low or loss of flow in passes	Tube failure, $1,000,000 loss	Med	Med	2
		Furnace fire, $1,000,000 loss	Med	Med	
2	Low pilot gas pressure	Explosion, multiple fatalities	Hi	Med	3

Note that a SIL was not selected for the manual shutdown. By definition, safety instrumented functions are automated systems that place the equipment or process in a safe state when certain conditions are violated. Alarms and manual shutdowns are usually *not* considered safety instrumented functions since they do not take automatic action. Therefore, many companies do not require SIL selections for alarm and manual shutdowns.

16.6 Develop Safety Requirement Specification (SRS)

Chapter 5 covered the safety requirement specification (SRS), which consists of the functional specification (what the system does), and the integrity specification (how well it does it). The SRS may in reality be a series of documents. Table 16-3 summarizes the key information that should be included in the SRS. Listing all of the requirements in a single table enables a simple crosscheck of what needs to be provided. The "details" column identifies whether information pertaining to each item should be provided for the application. If required, any special comments can be included in this column.

Table 16-3: SRS Summary Table

Item	Details of Requirement
Input Documentation and General Requirements	
P&ID's	Required. Figure 16-2 is a simplification of the actual P&ID.
Cause and effect diagram	Attached. (Refer to Table 16-4)
Logic diagrams	The cause and effect drawing is adequate to identify the logic requirements.

Process data sheets	Need to be provided for all field devices (i.e., FT-22, FT-23, PT-7, PT-24 and the four shutdown valves).
Process information (incident cause dynamics, final elements, etc.) of each potential hazardous event that requires a SIF.	Detailed descriptions are required on: - Details as to how explosions, fires, or tube failures can occur and how the protective instruments will mitigate these occurrences. - Speed of response and accuracy of the protective system needs to be defined. - Special requirements for the shutdown valves (e.g., fire safety, shutoff class).
Process common cause failure considerations such as corrosion, plugging, coating, etc.	The crude flow measurement is a difficult application due to high viscosity and freeze up. An in-line device is preferable. For this application the site has had good experience with Coriolis type meters. Corrosion due to hydrogen sulfide in the atmosphere needs to be addressed.
Regulatory requirements impacting the SIS.	The requirements of ANSI/ISA-84.00.01-2004, Parts 1-3 are to be satisfied. API 556 can be used as a reference. Full compliance with the standard is not mandatory.
Other	None
Detailed requirements for each SIF	
ID number of SIF	Refer to Table 16-4.
Required SIL for SIF	Refer to Table 16-4.
Expected demand rate	Once per year. Low demand operation.
Test interval	3 months
The definition of the safe state of the process, for each of the identified events.	The safe state is to shutoff the fuel gas, pilot gas, and the feed to the furnace.
The process inputs to the SIS and their trip points.	Refer to Table 16-4.
The normal operating range of the process variables and their operating limits.	Refer to Table 16-4.
The process outputs from the SIS and their actions.	Refer to Table 16-4.
The functional relationship between process inputs and outputs, including logic, math functions, and any required permissives.	Refer to Table 16-4.
Selection of de-energized to trip or energized to trip.	The complete safety system shall be de-energized to trip.

Consideration for manual shutdown.	One hardwired shutdown switch to be located in the main control room (HS-21). The manual shutdown must be independent of the programmable system.
Requirements related to the procedures for starting up and restarting the SIS.	Required. To be developed.
List of any special application software safety requirements.	None required.
Requirements for overrides or bypasses including how they will be cleared.	Required. To be developed.
Action(s) to be taken on loss of energy source(s) to the SIS.	All trip valves to close.
Response time requirement for the SIS to bring the process to a safe state.	5 seconds is acceptable.
Response action to diagnostics and any revealed fault.	Immediate response. Maintenance to be assigned highest priority. Closer monitoring of critical parameters must be put in place by operations until the problem is rectified.
Human-machine interfaces requirements.	Dedicated hardwired alarms in the main control room are required for any safety system fault or trip condition. Manual shutdown button required.
Reset function(s).	The trip valves shall be provided with manual reset solenoids.
Requirements for diagnostics to achieve the required SIL.	Smart field sensors are to be provided in order to utilize their diagnostic capabilities. Shutdown valves are to be provided with limit switches to verify that they have closed when requested by the logic system.
Requirements for maintenance and testing to achieve the required SIL.	Refer to the manual testing procedures.
Reliability requirements if spurious trips may be hazardous.	The spurious trip rate is to be calculated for compliance with the safety study requirements.
Failure mode of each control valve.	Fail closed.
Failure mode of all sensors/transmitters	Fail low.
Other.	None.

Table 16-4: Cause and Effect Diagram (Refer to Figure 16-3)

Tag #	Description	SIL	Instrument Range	Shut down Value	Units	Close Valve XV-30A	Close Valve XV-30B	Close Valve XV-31A	Close Valve XV-31B	Notes
FT-22	Pass flow A to furnace	2	0-500	100.0	B/H	X	X			1
FT-23	Pass flow B to furnace	2	0-500	100.0	B/H	X	X			1
PT-7/24	Pilot gas pressure	3	0-30	5	PSIG	X	X	X	X	2
	Loss of control power	NA				X	X	X	X	
	Loss of instrument air	NA				X	X	X	X	
HS-21	Manual Shutdown	NA				X	X	X	X	
Note 1: 20 second delay required before closing valves. Note 2: Solenoids with manual reset to be provided for all four shutdown valves.										

16.7 SIS Conceptual Design

The conceptual design must comply with any relevant company standards. A summary of the company guidelines relating to SIL and hardware selection is:

Table 16-5: SIS Design Guidelines Based on SIL

SIL	Sensors	Logic Solver	Final Elements
3	Redundant sensors required, either 1oo2 or 2oo3 depending on spurious trip requirements.	Redundant safety PLC required.	1oo2 voting required.
2	Redundancy may or may not be required. Initial option is not to have redundancy. Select redundancy if warranted by PFD_{avg} calculations.	Safety PLC required.	Redundancy may or may not be required. Initial option is not to have redundancy. Select redundancy if warranted by PFD_{avg} calculations.
1	Single sensor	Non-redundant PLC or relay logic.	Single device.

Based on the criteria in Table 16-5, the proposed system is shown in Figure 16-3.

Conceptual Design Requirements

The conceptual design (see Chapter 10) builds on and supplements the safety requirement specification (SRS) covered in Section 16.6 and incorporates the company's design guidelines. Key information required by the engineering contractor in order to complete the detailed engineering package should be provided. The design should adhere to the company standards and procedures. There should be no contradiction between the SRS and the conceptual design requirements.

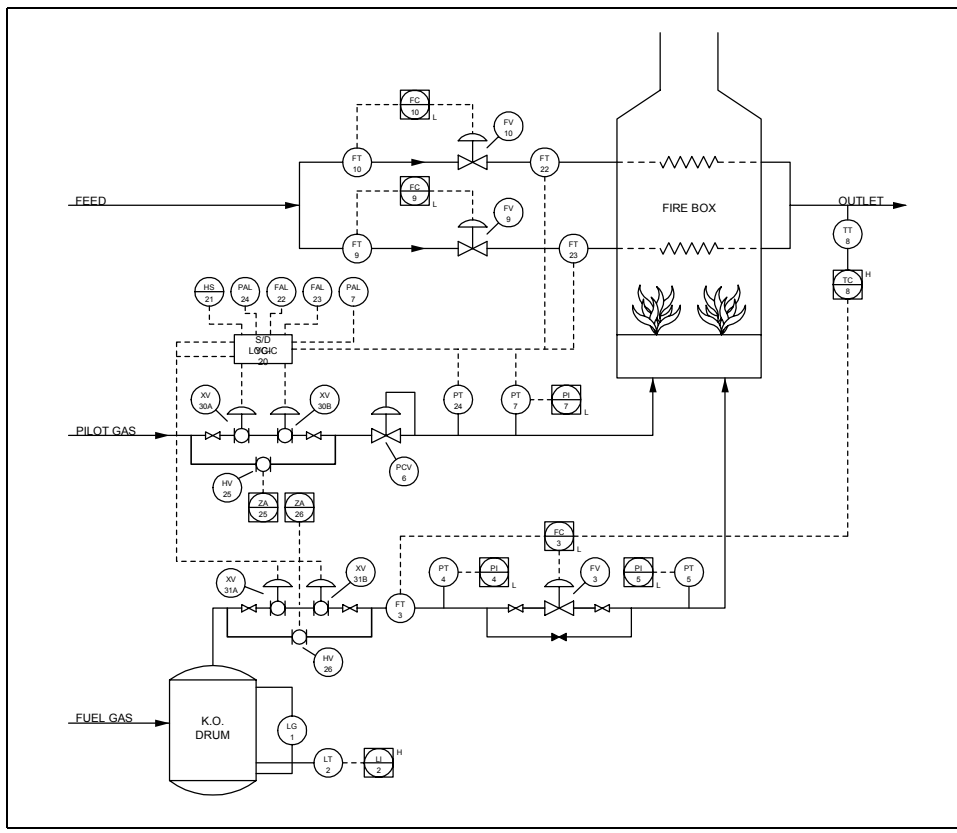


Figure 16-3: Proposed Safety Instrumented Functions

Table 16-6 summarizes the basic conceptual design requirements.

Table 16-6: Conceptual Design Summary

System architecture:	The logic solver shall be a redundant, certified safety PLC. The cabinet shall be located in the main control building. 1oo2 voting is required for the pilot gas pressure transmitters, as well as the fuel gas and pilot gas shutdown valves. Verification is required that this design will meet the SIL requirements. Figure 16-4 is a sketch of the overall configuration.
Minimize common cause:	Wiring from the 1oo2 valves and transmitters to the SIS to be segregated from BPCS wiring. Separate uninterruptible power supply (UPS) to power the SIS. All transmitters to have separate taps.
Environmental considerations:	The area classification is Class 1, Group D, Div 2. Hydrogen sulfide gas is in the environment around the furnace. The ambient temperature can fall to -35°C during the winter.
Power supplies:	110v, 60hz power is available from two separate UPS systems located in the main control room.
Grounding:	Ensure that company grounding standards for instrument and power systems are followed.
Bypasses:	Bypass valves are required to be installed around the pair of trip valves for the pilot gas and fuel gas trip valves for on-line testing. An alarm in the BPCS is required to indicate that a bypass valve has opened. No other bypasses are required.
Application software:	Ladder logic to be used for all programming in the SIS.
Security:	The existing company security requirements for access to and modification of the SIS logic shall be followed.
Operators interface:	Shutdown and diagnostic alarms are to be wired to an existing hardwired annunciator. The bypass alarms are to be connected to the BPCS.

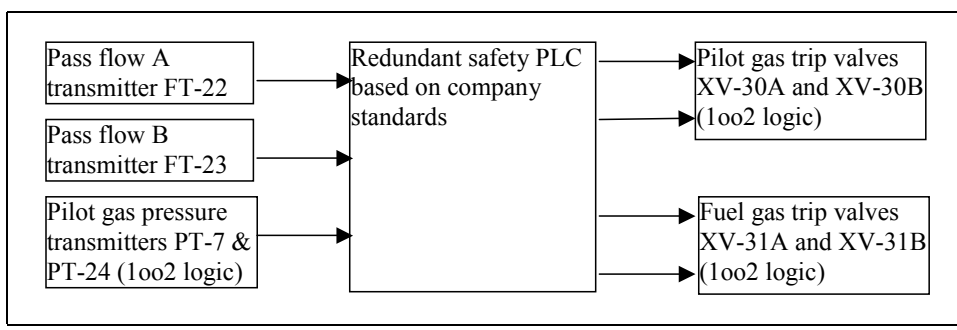


Figure 16-4: Proposed Conceptual SIS Design

16.8 Lifecycle Cost Analysis

Table 16-7: Lifecycle Cost Summary

Lifecycle Costs (20 years)	Material $	Labor $	Total cost $	Subtotal $
Initial Fixed Costs				
Safety classification		$1,000	$1,000	
SRS/Design specifications		$3,000	$3,000	
Detailed design and engineering		$20,000	$20,000	
Sensors	$24,000		$24,000	
Final elements	$6,000		$6,000	
Logic system	$30,000		$30,000	
Misc. – Power, wiring, jb's	$4,000		$4,000	
Initial training		$5,000	$5,000	
FAT/Installation/PSAT	$4,000	$16,000	$20,000	
Startup and correction	$1,000	$2,000	$3,000	
Fixed costs subtotal				$116,000
Annual Costs				
Ongoing training		$1,000	$1,000	
Engineering changes	$1,000	$1,000	$2,000	
Service agreement		$1,000	$1,000	
Fixed operation and maintenance costs		$1,000	$1,000	
Spares	$4,000		$4,000	
Online testing		$8000	$8,000	
Repair costs	$1,000	$500	$1,500	
Hazard costs				
Spurious trips costs			$8,000	
Annual costs subtotal				$26,500
Present value for annual costs (20 years, 5% interest rate)				$330,249
Total Lifecycle Costs				**$446,249**

16.9 Verify that the Conceptual Design Meets the SIL

It's important to verify that each safety instrumented function (SIF) meets the safety integrity level (SIL) requirements. In this case we have three functions (i.e., low pass flow A, low pass flow B, and low pilot gas pressure). The low pilot gas pressure function requires SIL 3 and the pass flow functions require SIL 2. Only the SIL 3 function will be analyzed here.

The equations outlined in Chapter 8 are used for the calculations. A block diagram of the conceptual pilot gas shutdown function is shown in Figure 16-5.

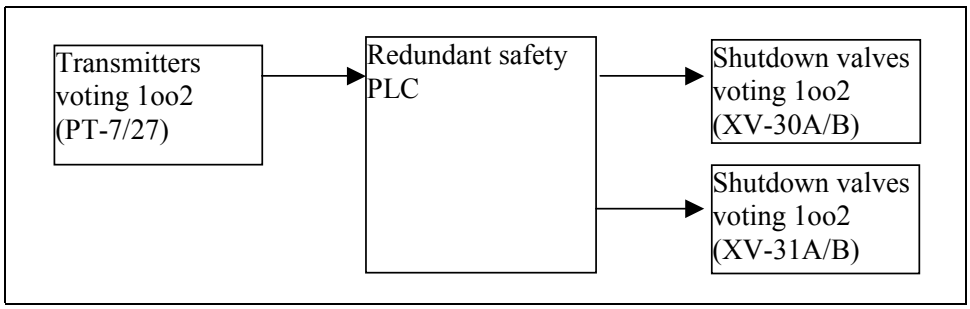


Figure 16-5: Block Diagram for Pilot Gas Shutdown Function

Assuming:

Mean time to repair (MTTR):	8 hours
Average demand rate:	Once per year
Manual test interval :	3 months
Common cause β factor:	5%
Transmitter diagnostics using comparison:	90%

Table 16-8: Failure Rate Data (Failures per Year)

Item	Dangerous failure rate λ_d	Safe failure rate λ_s
Transmitter PT-7/24	0.01 (1/100 years)	0.02 (1/50 years)
Valve and solenoid XV-30A/B, XV-31A/B	0.02 (1/50 years)	0.1 (1/10 years)
Safety PLC	See note 1	See note 1
Note 1: The PFD_{avg} and MTTFsp values for the redundant safety PLC are supplied by the vendor.		

PFD_{avg} Calculations:

PFD_{avg} (Sensors) = 0.01 * 0.1 * 0.05 * 0.25/2 = 0.00000625
(failure rate * 10% undetected failures * 5% common cause * test interval/2)
PFD_{avg} (Valve and solenoid) = 2 * 0.02 * 0.05 * 0.25/2 = 0.00025
(quantity * failure rate * 5% common cause * test interval/2)
Redundant safety PLC: = = 0.00005
PFD_{avg} (Total) = 0.000306

The maximum value allowed for SIL 3 is 0.001, therefore the conceptual design satisfies the safety requirements. The risk reduction factor (RRF = 1/PFD) for the system is 3,300, which is between the range of 1,000 and 10,000 for SIL 3.

$MTTF^{spurious}$ Calculations:

All field devices are included in nuisance trip calculations.

$MTTF^{spurious}$ (Sensors) = 1/(4 * 0.02) = 12.5 yrs
$MTTF^{spurious}$ (Valve and Solenoid) = 1/(4 * 0.1) = 2.5 yrs.
MTBFsp (Safety PLC) = 1/(0.01) = 100 yrs.
MTBFsp (Total) = 2.0 years

A nuisance trip is expected to occur, on average, every two years, which also meets the original requirements.

16.10 Detailed Design

The detailed design needs to comply with the safety requirement specification and the conceptual design. The following documentation shall be provided as part of the detailed design package.

1. results and recommendations of safety/hazard studies
2. all PFD_{avg} and $MTTF^{spurious}$ calculations
3. piping and instrument diagrams (P&IDs)
4. instrument index
5. specification sheets for safety PLC, transmitters, and valves
6. loop diagrams
7. cause and effect matrix
8. drawings to indicate locations of all major equipment and controls

9. PLC system configuration, I/O listings, and ladder program listings
10. junction box and cabinet connection diagrams
11. power panel schedules
12. pneumatic system tubing diagrams
13. spare parts list
14. vendor standard documentation including specifications, installation requirements, and operating/maintenance manuals
15. checkout forms and procedures
16. manual test procedures

16.11 Installation, Commissioning and Pre-startup Tests

The following activities need to be completed.

- Factory acceptance test (FAT) of logic system. The responsibilities and details of the test need to be included as part of the detail design activities. Support personnel shall witness the test.
- Field installation of the complete system. The field installation must conform to the installation drawings included in the design package.
- Installation checks. Device checkout forms must be completed and included with the installation scope of work.
- Validation or site acceptance tests. Functional checkout per the functional checkout forms and procedures.
- Pre-startup acceptance test (PSAT). To be completed by operations personnel prior to startup of the unit. The cause and effect diagram will be adequate for the PSAT.

16.12 Operation and Maintenance Procedures

Procedures must be developed for safe operation and maintenance of the unit. Test procedures must be developed for testing each safety instrumented function. The following procedure covers testing of the low pilot gas pressure shutdown function.

Shutdown Test Procedure Procedure # XX-1

1.0 Purpose
To test the low pilot gas pressure trips for furnace F-001 and transmitters PT-7 and PT-24.

2.0 Responsibility for test
The furnace operator is responsible for completing the test. The resident instrument technician shall accompany the operator during the test.

3.0 Tools and other items required
Portable radio. No special tools are required to complete this test.

4.0 Trip check frequency
This test has to be completed every 3 months

5.0 Hazards
Failure to open the bypass valves for the pilot gas or the fuel gas valves will trip the furnace and can create a furnace fire or explosion.

Failure to close the bypass valves fully upon completing the test can disable the furnace shutdown functions.

6.0 Reference information
The following documentation should be readily available to assist in rectifying any problems: loop drawings, instrument spec sheets, and shutdown system description.

7.0 Detailed step-by-step test procedure
Refer to attached sketch (Figure 16-6) with trip system description.

Step #	Action	Check OK
1	Advise all operating unit personnel that the test is about to commence.	
2	Open pilot gas bypass valve HV-25 fully. Verify bypass alarm ZA-25 in BPCS. Acknowledge the alarm.	
3	Open fuel gas bypass valve HV-26 fully. Verify bypass alarm ZA-26 in BPCS. Acknowledge the alarm.	
4	Get confirmation that the furnace operation is still stable.	
5	Close process isolation valve for PT-7. Slowly vent the pressure to the transmitter. The two pilot gas trip valves and two fuel gas trip valves should trip at 5 psig. Record trip pressure. Verify that all four valves have closed fully. Verify the speed of operation is < 3 seconds.	

6	Verify that alarm PAL-7 was activated in the main control room annunciator and in the BPCS. Acknowledge the alarm.	
7	Re-pressurize PT-7.	
8	Alarm PAL-7 should clear. Verify	
9	Reset the solenoids for the four trip valves. The valves should reopen instantly. Verify.	
10	Repeat steps 5 through 9 for PT-24.	
11	Close pilot gas bypass valve HV-25 fully. Verify that alarm ZA-25 in the BPCS cleared.	
12	Close fuel gas bypass valve HV-26 fully. Verify that alarm ZA-25 in the BPCS cleared.	
13	Advise all operating personnel that the trip test for PT-7 and PT-24 is complete.	
14	Complete, sign, and forward the completed documentation to plant engineer for review and filing.	

Results/Comments/Deficiencies:

Shutdown Check Completed By:_____ Date: _____

Operation: If either PT-7 or PT-24 pressure drops below 5 PSIG valves XV-30A, XV-30B, XV-31A and XV-31B will close within 3 seconds.

Summary

The fired heater case study in this chapter shows how the material and techniques outlined in Chapters 1 through 15 can be used to specify, design, install, commission, and maintain atypical safety instrumented system. The following sections of the lifecycle model were addressed for the application:

- define the target SIL for each SIF
- develop the safety requirement specification (SRS)
- SIS conceptual design
- lifecycle cost analysis
- verify that the conceptual design meets the SIL
- detail design
- installation, commissioning and pre-startup tests
- operation and maintenance procedures

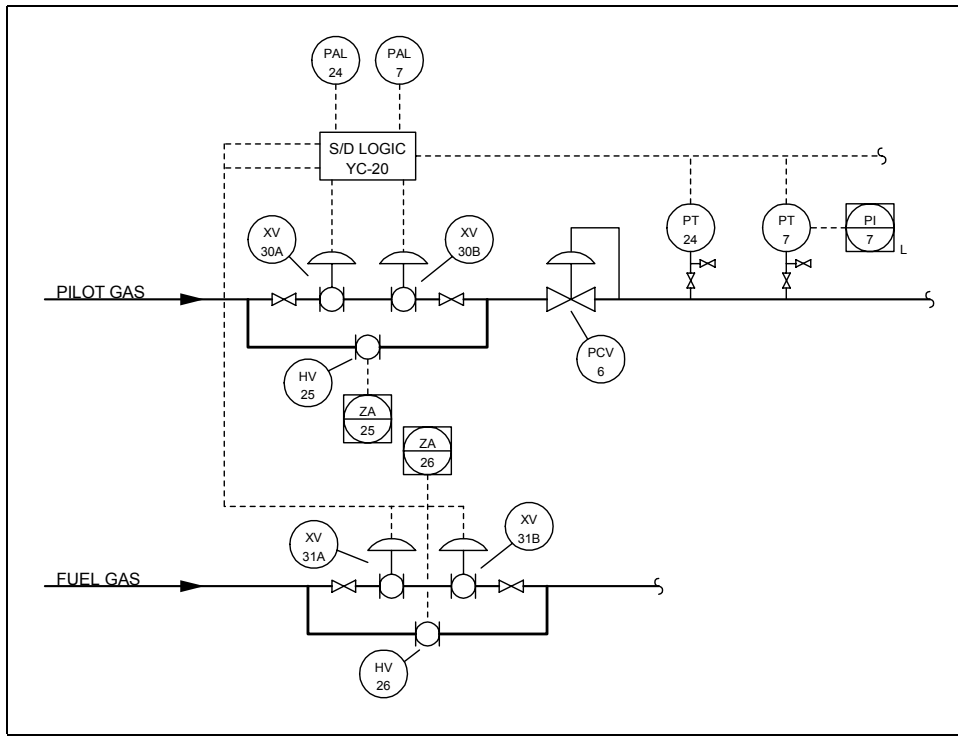


Figure 16-6: Low Pilot Gas Pressure Trip System Diagram and Description

References

1. ANSI/ISA-84.00.01-2004, Parts 1-3 (IEC 61511-1 to 3 Mod). *Functional Safety: Safety Instrumented Systems for the Process Industry Sector.*